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University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Autumn 2025

Course Title: Digital Logic Design

Course Code: CSE 0611215/CSE215

Marks: 50

Time: 3(three)hours

(Answer all questions)

Q. No.

Marks

1. a) A device measures two physiological signals—heart-rate correction value and oxygen-level adjustment value—each represented as a 6-bit binary number. To combine these signals in real time, the processor uses a 6-bit parallel adder constructed from 3-bit full-adder blocks connected in cascade. During one measurement cycle, the device produces the following readings:

[05]

- $A = 101011_2$
- $B = 011110_2$
- $Cin = 1$.

Draw the block diagram showing how 3-bit adders are connected to build the 6-bit parallel adder and **compute** the sum and final carry-out.

- b) A digital circuit designer is building a low-power arithmetic unit for an embedded security sensor. To minimize hardware complexity, the designer decides to implement the full subtractor using only NOR gates, since NOR is a universal gate.

[05]

Now **construct** the gate-level schematic diagram of the full subtractor implemented solely with NOR gates with a proper truth table.

2. a) A small toy counter produces a 4-bit binary number (0–9) but the next stage requires Excess-3 code to drive another module. **Design** a circuit that converts a 4-bit binary input into its 4-bit Excess-3 equivalent.

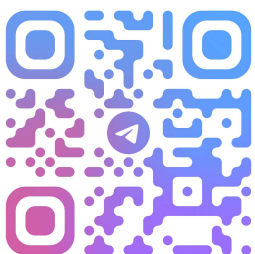
[06]

- b) **Implement** a 5-to-32-line decoder circuit by interconnecting one 2-to-4 decoder and four 3-to-8 decoders with enable inputs.

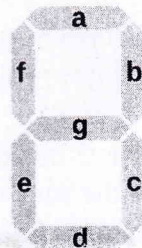
[04]

3. a) An 8-to-3 encoder circuit that converts eight input lines into a 3-bit binary output. The encoder has eight input signals (D_0 – D_7), where only one input is active at a time, and three output lines (Y_2 , Y_1 , Y_0) representing the binary code corresponding to the active input. **Construct** the truth table, derive the simplified Boolean expressions for each output line and draw the logic diagram of the encoder.

[05]



- b) To design a full adder circuit using two half adders and an OR gate, you need to complete the following task. [05]
- Draw the logic diagram of a full adder.
 - Write the truth table, and
 - Derive the Boolean expressions for the Sum and Carry outputs.
 - Finally, provide a **proof** showing that the designed circuit performs the same logic function as a standard full adder.
4. a) **Evaluate** the following boolean function using a 4x1 MUX. [05]
- $$F(P, Q, R, S) = \Sigma(0, 1, 4, 5, 10, 11, 12, 13, 15)$$
- b) **Design** a JK Flip Flop by using a D Flip Flop. [05]
5. a) A seven-segment display is used to represent the word APPLE. [10]



Find the logical expression for segment 'e' using a truth table and K-map simplification. The BCD value for APPLE will be 0,1,2,3,4.

